

# When Sample Selection Bias Precipitates Model Collapse

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## Introduction

Recursive synthetic-data training can narrow distributional support when later generations are selected by biased local references. The paper studies when low-resource verification regimes lose tail modes before accumulation can preserve them.

## Theory

**Replace paradigm [R1]**  
Replacing data by the latest generated round drives covariance loss and tail forgetting.

**Accumulate paradigm [R2]**  
Pooling real and synthetic generations can stabilize unbiased recursion, but local selection changes the limiting distribution.

**Theorem 1**  
Top-rank selection toward a local reference moves the mean toward that reference while covariance contracts.

**Theorem 2**  
Within the local basin, the covariance trace follows a power-law decay; thinner support reaches the collapse signal earlier.

**Theorem 3**  
After fitting error is small, target performance is governed by Wasserstein mismatch to the true manifold.

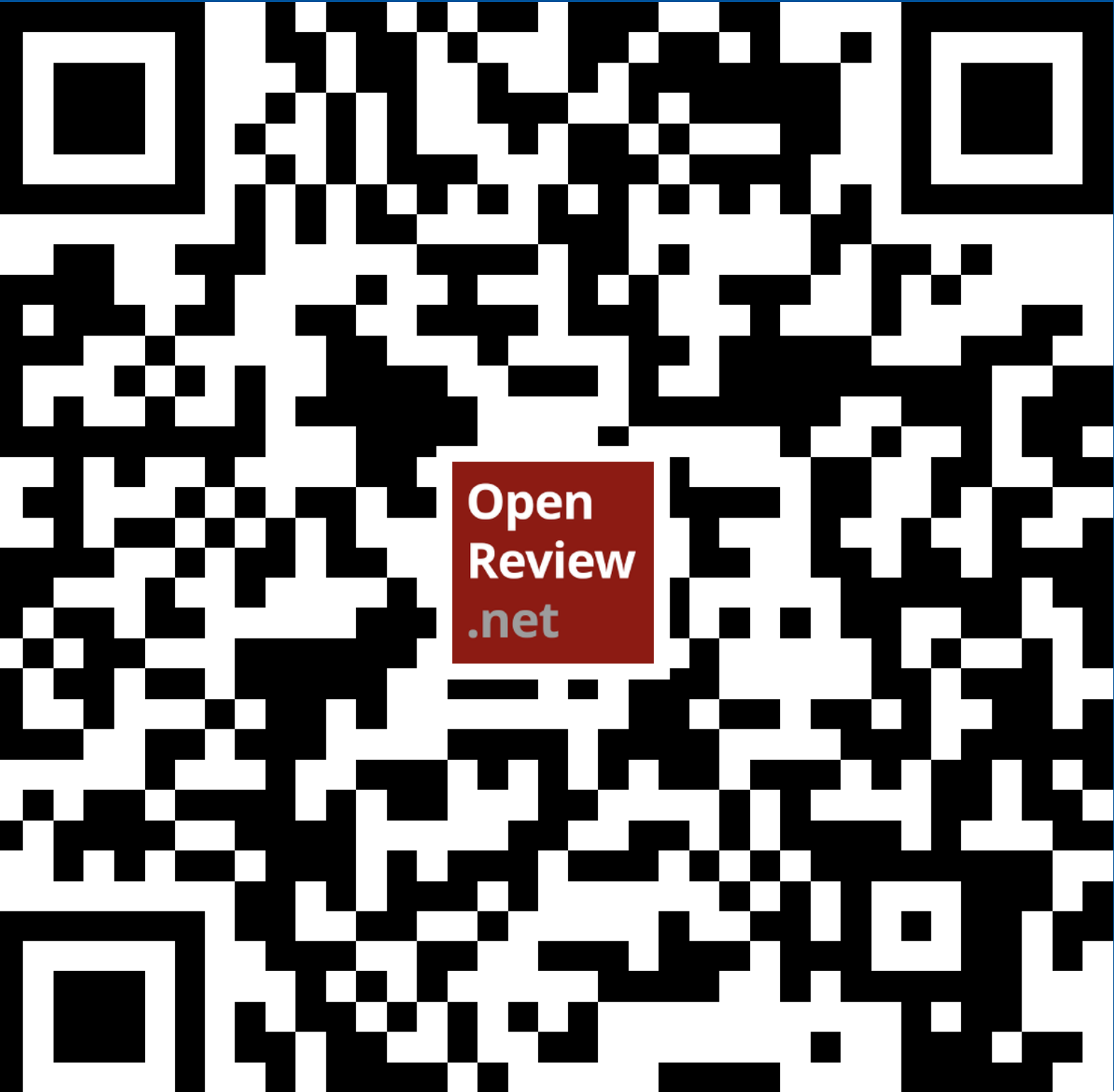
## References

[R1] Shumailov et al., *Nature*, 2024.  
[R2] Gerstgrasser et al., arXiv:2404.01413, 2024.

For more methodology and theoretical details, please refer to the full paper.



Selection bias  
prunes rare modes.  
Low-resource silos  
collapse earlier.  
Tail coverage  
decays persistently.



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## Multivariate Gaussian Modeling

**Gaussian.** Closed-form  $W_2$  dynamics separate mean drift from covariance shrinkage and make dispersion loss directly measurable.

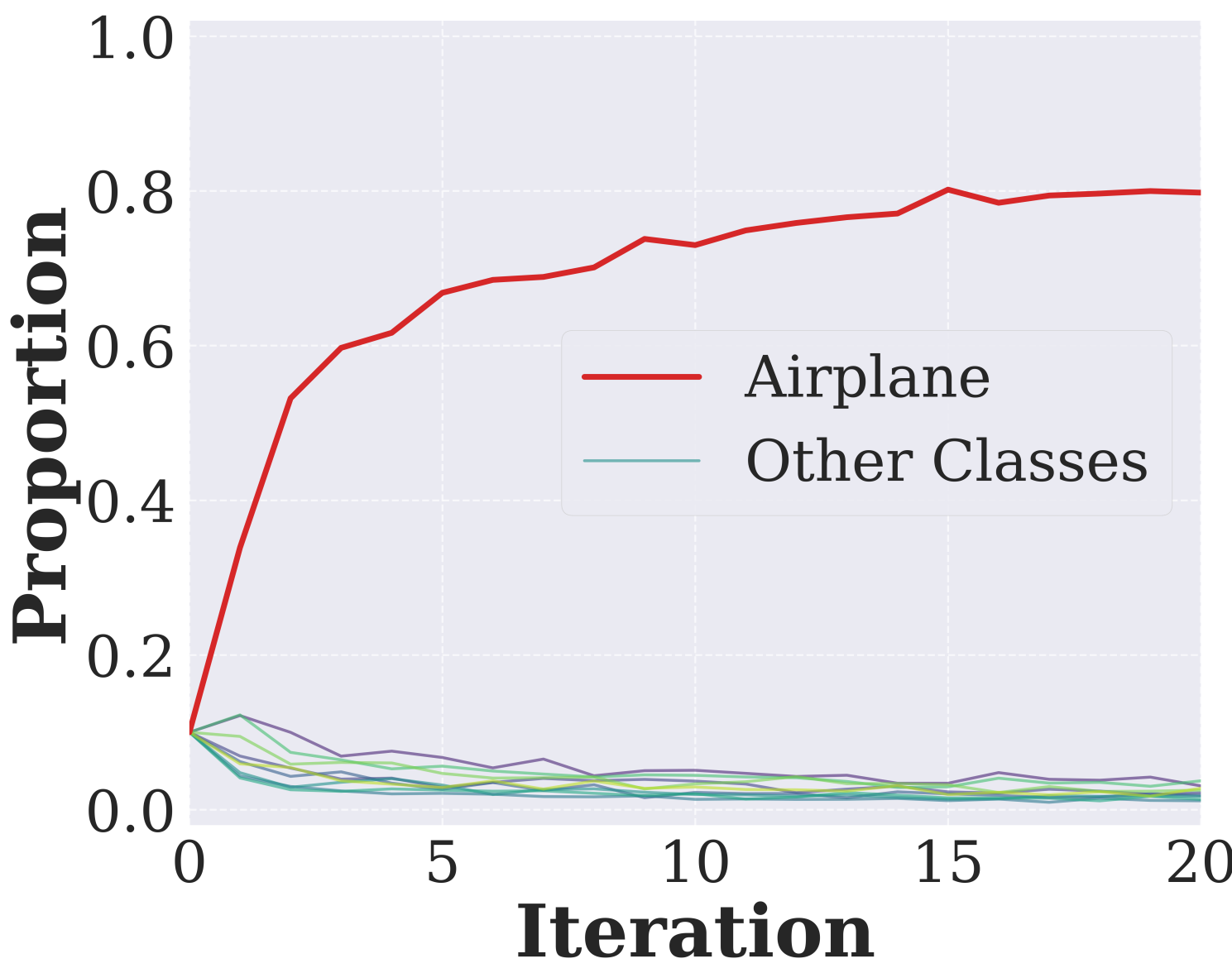
**Anisotropic Gaussian**  
Directional variance changes the path, but not the loss of dispersion.

**Gaussian mixture**  
Biased filtering reduces mode coverage across generations.

**Laplace**  
Heavy-tailed initialization still loses dispersion under biased filtering.

## Collapse Signal

**Trace-ratio diagnostic**  
The trace ratio summarizes diversity contraction and exposes persistent tail pruning.



## Empirical Summary

Across Gaussian and non-Gaussian families, biased selection is the dominant driver of diversity loss and reduced coverage.

For additional experimental results, please refer to the full paper.

